Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18CS5CM03]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PRINCIPLES OF VIRTUALIZATION

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Describe the cloud service model and justify how these models will be beneficial for current IT organizations?	**08**
	b.	What is a Virtual machine? Write down the basic steps to create and configure the VM.	**06**
2.	**a.**	Differentiate DAS, NAS and SAN technologies.	**07**
	b.	With the help of a neat diagram of Virtualization architecture, explain the role of hypervisor in Virtualization. Also describe the types of hypervisors.	**07**
3.	**a.**	Write a short note on – (i) Network Adapter (ii) Port Group and Uplink	**06**
	b.	Exemplify the terms CAPEX(Capital Expenditures) and OPEX (Operational Expenditures) from the perspective of IT organizations and how virtualization will help to reduce CAPEX and OPEX.	**08**
		OR	
4.	**a.**	Explain the term HA (High Availability) and how it will be benefit to the IT organizations? Also, differentiate between HA and FT (Fault Tolerance).	**07**
	b.	Differentiate between DAS, NAS, and SAN. Which one is better? Support your answer with the help of a case study.	**07**
5.	**a.**	Differentiate between HOST, CLUSTER, and RESOURCE POOLING.	**06**
	b.	With the help of a case study, Elaborate the concept of Load Balancing - DRS and DPM in the virtual IT Infrastructure.	**08**
		OR	
6.	**a.**	Discuss the brief introduction for Network virtualization, and With the help of the diagram, Elaborate anyone from the following: VSS (Virtual Standard Switch) and vDS(Virtual distributed switch).	**06**
	b.	Currently, with the growth of large amounts of data, Data Centers are facing many challenges in the present IT trend such as Scalability, Availability, and Management challenges. Discuss in brief any two challenges mentioned above.	**08**

7.	**a.**	Write the steps to create and configure of ESXi Host in Type 2 Hypervisor. List Down the features of the ESXi host.	**07**
	b.	Write the comparative analysis between the mentioned virtualization tools: VMware Vsphere and Microsoft Hyper V along with the case study.	**07**

OR

8.	**a.**	Give a brief introduction to Vcenter Server and list down all the features of the Vcenter Server.	**07**
	b.	Illustrate the current challenges society faces, which can be addressed by virtualization technology.	**07**